

SIERRA ROBISON

sierra.liangshu@gmail.com | sierrar@bu.edu | www.linkedin.com/in/sierra-robison

EDUCATION

Boston University, College of Engineering — B.S. Biomedical Engineering (Expected May 2027)

GPA: 3.84 | **Dean's List:** 2023 - Present

Relevant Courses: Physics 1&2, Static Mechanics, Differential Equations, Linear Algebra, Probability & Statistics & Data Science, MATLAB, Electric Circuits, Hands-On-Engineering, Biomedical Instrumentation, Mechanobiology, Thermodynamics, Signals & Controls, Biomedical Measurements

WORK EXPERIENCE

Boston University Engineering Product Innovation Center

Spring 2025 - Present

Lab Assistant

- Assisting in operation of fabrication equipment, providing troubleshooting for a variety of prototyping projects, and improving the lab space through in-house projects [6 hours] weekly during the school semester and [30 hours] weekly during the summer

Boston University College of Mechanical Engineering

Spring 2025

Static Mechanics Learning Assistant

- Worked [8 hours] per week, assisting students with problem solving and conceptual understanding during class and office hours

Medtronic

Summer 2024

Research & Technology Engineering Intern

- Contributed [40 hours] per week working within the Surgical Robotics Research and Technology team as a WISE Intern
- Tested novel 6D pose estimation (SO-Pose) computer vision approach for the HUGO Robotic-Assisted Surgery machine

Boston University Admissions

Spring 2024 - Present

Admissions Ambassador and Evaluator

- Providing a welcoming environment to prospective students and evaluating team members to ensure positive experiences

PROJECTS

Engineering Product Innovation Center Projects

Summer 2025

- Introductory projects involving manual and CNC mills and lathes, laser cut topographical map demonstrating the versatility in functionality of laser cutters, prototyping demonstration using 3D scanner, and training and organizational lab materials

Gait Tracker

Fall 2024

- Collaborated with a small team in a semester-long project to create a device utilizing Bluetooth to record heel and toe strikes of stroke therapy patients in BU's Neuromotive Recovery Lab in order for medical professionals to monitor patient recovery

Medtronic 6D Pose Estimation

Summer 2024

- Enhanced the HUGO system's ability to infer hidden geometry from 2D data by extending the original GDR-Net training code to support SO-Pose, which incorporates self-occlusion reasoning in addition to 2D-3D point correspondences

AI4ALL COVID Diagnostic

Summer 2022

- Collaborated with peers and a mentor to train a computer vision model to differentiate x-rays of lungs with/without COVID-19

EXTRACURRICULAR ACTIVITIES

BU College of Engineering

Spring 2024 - Present

Student Advisor

- Guiding engineering students through their first semester at BU by providing social, academic, and professional support

Beta Psi Omega

Fall 2023 - Present

- Working alongside other members to gain professional skills
- Participating in weekly meetings and additional events

SKILLS

Engineering: OnShape, SolidWorks, manual mill, manual lathe, CNC mill, CNC lathe, 3D printers (Prusa and Bambu), laser cutter, waterjet, soldering

Wet Lab: Gel electrophoresis, PCR, DNA barcoding

Language: Python, MATLAB, Arduino IED, Java, HTML

Application: Google, Microsoft, Jira, Linux, Bitbucket